

UNIT 4 • LECTURE NOTES

Unit IV – Ideal and Real Gases, Thermodynamic Relations

Detailed lecture notes — ideal-gas laws and properties, real gases and the compressibility factor, equations of state, corresponding states and the generalised compressibility chart, Maxwell relations, Tds equations, specific-heat relations, energy equation, Joule–Thomson coefficient and the Clapeyron equation, with solved numericals

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SYLLABUS

UNIT IV — IDEAL AND REAL GASES, THERMODYNAMIC RELATIONS (9 + 3 periods)

Properties of Ideal gas – Ideal and real gas comparison – Equations of state for ideal and real gases – Reduced properties – Compressibility factor – Principle of Corresponding states. Generalised Compressibility Chart and its use. Maxwell relations, Tds Equations, Difference and ratio of heat capacities, Energy equation, Joule-Thomson Coefficient, Clausius Clapeyron equation, Phase Change Processes. Simple Calculations.

Course Outcome CO4: Solve the practical problems by understanding the behavior of Ideal gas and the interrelationship between thermodynamic functions.

Text: P. K. Nag, *Engineering Thermodynamics*, 6th Edn., McGraw-Hill (Chapters 10 and 11).

References: Cengel & Boles, *Thermodynamics – An Engineering Approach*, 8th Edn.; R. K. Rajput, *A Text Book of Engineering Thermodynamics*, 5th Edn. Real-gas reference data in these notes are from IAPWS-97 (steam) and the NIST reference equations (nitrogen, carbon dioxide).

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Learning objectives. After studying this unit you should be able to: (i) state the gas laws and the characteristic gas equation, distinguish the universal and characteristic gas constants and list the assumptions of the ideal-gas model; (ii) define the compressibility factor and explain, with the help of Z - p diagrams, how and why real gases deviate from ideal behaviour and when a real gas may be treated as ideal; (iii) derive the van der Waals equation, obtain its constants from the critical properties, and write and compare the Redlich–Kwong, Beattie–Bridgeman and virial equations; (iv) define reduced properties, state the principle of corresponding states and use the generalised compressibility chart to find p , v or T of a real gas; (v) compare the behaviour of air and steam in constant-volume and constant-pressure processes; (vi) define the thermodynamic potentials, derive the four Maxwell relations and the two Tds equations; (vii) derive and apply the expressions for $c_p - c_v$, c_p/c_v , the energy equation, the Joule–Thomson coefficient and the Clapeyron and Clausius–Clapeyron equations to gases, liquids and phase changes.

1. Introduction – why real gases need more than $p v = RT$

Units I and II treated gases with the ideal-gas equation $p v = RT$, and Unit III showed that steam near its saturation line cannot be treated in this way – its properties had to be tabulated. Between these extremes lie the real gases of engineering practice: air in a high-pressure compressor, nitrogen or oxygen in a storage cylinder, refrigerants, natural gas in a pipeline, carbon dioxide in a supercritical cycle. At high pressure or near the critical temperature these gases deviate appreciably from $p v = RT$ because their molecules occupy a finite volume and attract one another. This unit has two parts. The first (Sections 2 to 6) describes the ideal gas, measures the deviation of real gases by the compressibility factor, and gives the two practical tools for real gases – equations of state such as van der Waals' and the generalised compressibility chart based on the principle of corresponding states. The second part (Sections 7 to 10) develops the **general thermodynamic relations** – the Maxwell relations, the Tds equations, the expressions for $c_p - c_v$ and c_p/c_v , the energy equation, the Joule–Thomson coefficient and the Clapeyron equation – by which properties that cannot be measured directly (entropy, internal energy, enthalpy) are calculated from those that can (p , v , T , c_p , compressibility, expansivity).

2. The ideal gas – gas laws, equation of state and properties

2.1 The gas laws

- **Boyle's law (1662):** at constant temperature the volume of a given mass of gas is inversely proportional to its pressure, $pV = \text{constant}$ ($T = C$), so $p_1 V_1 = p_2 V_2$.
- **Charles' law (1787):** at constant pressure the volume of a given mass of gas is directly proportional to its absolute temperature, $V/T = \text{constant}$ ($p = C$); equivalently, at constant volume $p/T = \text{constant}$ (sometimes called Gay-Lussac's law).
- **Avogadro's law (1811):** equal volumes of all gases at the same pressure and temperature contain the same number of molecules; hence the volume of one kilomole of any gas at a given p and T is the same – 22.414 m^3 at 0°C and 101.325 kPa – and one kilomole contains 6.022×10^{26} molecules.

2.2 The characteristic gas equation

Combining Boyle's and Charles' laws for a fixed mass of gas gives $pV/T = \text{constant}$. For 1 kg the constant is the **characteristic (specific) gas constant** R of the gas, and for m kg,

$$p v = RT \quad p V = m R T \quad (p \text{ in kPa, } v \text{ in m}^3/\text{kg, } R \text{ in kJ/kg K, } T \text{ in K})$$

By Avogadro's law the volume of one kilomole is the same for all gases, so on a molar basis the constant is the same for all gases – the **universal gas constant** $R_u = 8.3145 \text{ kJ/kmol K}$ ($= 8314.5 \text{ J/kmol K}$). With $n = m/M$ kilomoles ($M = \text{molar mass, kg/kmol}$) and molar volume $\bar{v} = V/n$,

$$pV = nR_u T \quad p\bar{v} = R_u T \quad R = R_u/M$$

Thus R_u is the same for every gas, while R is a property of the particular gas: $R_{\text{air}} = 8.3145/28.97 = 0.287 \text{ kJ/kg K}$, $R_{N_2} = 8.3145/28.013 = 0.2968$, $R_{O_2} = 0.2598$, $R_{CO_2} = 8.3145/44.01 = 0.1889$, $R_{H_2} = 4.124$, $R_{\text{steam}} = 8.3145/18.015 = 0.4615 \text{ kJ/kg K}$. The lighter the gas, the larger its R .

A short example. A gas of $R = 0.287 \text{ kJ/kg K}$ occupies 0.4 m^3 at 3 bar and 35°C . Its mass is $m = pV/(RT) = 300 \times 0.4/(0.287 \times 308.15) = 120/88.44 = 1.357 \text{ kg}$; the number of kilomoles is $n = m/M = 1.357/28.97 = 0.0468 \text{ kmol}$, and $pV = nR_u T = 0.0468 \times 8.3145 \times 308.15 = 120 \text{ kJ}$ ✓.

2.3 Definition and assumptions of the ideal gas

An **ideal (perfect) gas** is one that obeys $p\bar{v} = RT$ exactly at all pressures and temperatures. The kinetic theory derives this equation from the following assumptions: (i) the gas consists of a very large number of identical molecules in random motion; (ii) the volume of the molecules themselves is negligible compared with the volume of the gas; (iii) there are no forces of attraction or repulsion between the molecules except during collisions, so the molecules move in straight lines between collisions; (iv) collisions between molecules and with the walls are perfectly elastic and of negligible duration; (v) the molecules obey Newton's laws. Assumptions (ii) and (iii) are the ones that fail for real gases at high density. No real gas is ideal, but every gas approaches ideal behaviour as its pressure is reduced ($p \rightarrow 0$) and its temperature is raised well above the critical temperature, because the molecules are then far apart and their volume and mutual attraction become negligible. Air, nitrogen, oxygen, hydrogen, helium and the products of combustion at ordinary pressures and temperatures are treated as ideal gases with errors well under 1 %; steam in air-conditioning applications (partial pressure a few kPa) is likewise an ideal gas, but steam in a power plant is not.

2.4 Properties of an ideal gas

- **Joule's law:** the internal energy of an ideal gas depends on temperature alone, $u = u(T)$ – Joule's free-expansion experiment found no temperature change when a gas expanded into a vacuum, and the energy equation of Section 8.4 proves the result from $p\bar{v} = RT$. Hence $h = u + p\bar{v} = u + RT$ is also a function of temperature alone.
- **Specific heats:** $c_v = du/dT$ and $c_p = dh/dT$ are functions of temperature only; $du = c_v dT$ and $dh = c_p dT$ for any process of an ideal gas; $c_p - c_v = R$ and $\gamma = c_p/c_v$, so $c_v = R/(\gamma - 1)$ and $c_p = \gamma R/(\gamma - 1)$. For a perfect gas the specific heats are constant (air: $c_p = 1.005$, $c_v = 0.718 \text{ kJ/kg K}$, $\gamma = 1.4$); for a real gas at high temperature they increase with T .
- **Entropy:** $s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln(v_2/v_1) = c_p \ln(T_2/T_1) - R \ln(p_2/p_1)$ (Unit II).
- **Processes:** the polytropic relations $p\bar{v}^n = C$, $T\bar{v}^{n-1} = C$ and $T/p^{(n-1)/n} = C$, with $n = 0$ (isobaric), 1 (isothermal), γ (isentropic) and ∞ (isochoric), and the work and heat expressions of Unit I.

3. Real gases – the compressibility factor and the deviation from ideal behaviour

3.1 Ideal gas and real gas compared

Feature	Ideal gas	Real gas
Equation of state	$p\bar{v} = RT$ exactly at all states	$p\bar{v} = ZRT$, $Z \neq 1$; van der Waals, Redlich–Kwong, virial, etc.
Molecular volume	Negligible	Finite; important at high pressure ($Z > 1$)

Feature	Ideal gas	Real gas
Intermolecular forces	None	Attraction at moderate density ($Z < 1$), repulsion at very high density
u and h	Functions of T only; $c_p - c_v = R$	Functions of T and p (or v); $c_p - c_v \neq R$
Joule–Thomson coefficient	Zero	Non-zero; cools or warms on throttling
Liquefaction	Cannot liquefy	Liquefies below the critical temperature
Where valid	Low pressure, high temperature ($p_R \ll 1$ or $T_R > 2$)	All states; needed near the saturation region and the critical point

3.2 The compressibility factor

The deviation of a real gas from ideal behaviour is measured by the **compressibility factor**

$$Z = pv/(RT) = v_{\text{actual}}/v_{\text{ideal}} \quad (Z = 1 \text{ for an ideal gas})$$

Z is the ratio of the actual specific volume of the gas to the volume it would have as an ideal gas at the same p and T. Figure 4.1 (left) shows Z for nitrogen: at every temperature $Z \rightarrow 1$ as $p \rightarrow 0$; at low temperatures Z first falls below 1 with increasing pressure, because the attractive forces pull the molecules closer together and make the gas *more* compressible than an ideal gas, and then rises above 1 at high pressure, because the finite volume of the molecules leaves less free space and makes the gas *less* compressible. The dip is deepest near the critical temperature (Z can be as low as 0.2 near the critical point) and disappears at high temperatures, where the molecules move too fast for the attractions to matter. For a van der Waals gas the value at the critical point is $Z_c = 3/8 = 0.375$; real gases have Z_c between about 0.23 (water) and 0.31 (helium, hydrogen), most simple gases lying near 0.27–0.29.

3.3 Boyle temperature

The temperature at which the initial slope $(\partial Z/\partial p)_T$ at $p \rightarrow 0$ is zero – so that the gas obeys Boyle's law over an appreciable range of pressure – is the **Boyle temperature** T_B ; the attractive and repulsive effects then cancel at low density, and the second virial coefficient (Section 4.5) vanishes. Below T_B the isotherms of Z dip below 1; above it they rise monotonically. For a van der Waals gas $T_B = a/(Rb) = 27T_c/8$; the measured values are lower (nitrogen 327 K, hydrogen 110 K, helium 23 K).

3.4 When can a real gas be treated as ideal?

Z is within about 1 % of unity, and the ideal-gas equation is adequate, when the pressure is very low compared with the critical pressure ($p_R = p/p_c \ll 1$) at any temperature, or when the temperature is well above the critical temperature ($T_R = T/T_c > 2$) at any pressure short of very high values ($p_R \gg 1$). Air at atmospheric conditions ($T_R = 2.2$, $p_R = 0.03$) is ideal; steam in the atmosphere ($p_R = 0.0001$) is ideal; steam in a boiler at 10 MPa and 400 °C ($p_R = 0.45$, $T_R = 1.04$) is not ($Z = 0.84$).

4. Equations of state for real gases

An **equation of state** is any relation $f(p, v, T) = 0$ among the three measurable properties. For real gases it must reproduce the behaviour described above; many have been proposed, from the two-constant van der Waals equation to multi-constant empirical forms.

4.1 The van der Waals equation – derivation

Van der Waals (1873) corrected the ideal-gas equation for the two neglected molecular effects.

- **Finite molecular volume.** The molecules themselves occupy a volume, so the space available for molecular motion is not v but $(v - b)$, where b, the **co-volume**, is about four times the actual volume

of the molecules per unit mass. The pressure exerted by the gas is therefore larger than the ideal value for the same v : $p = RT/(v - b)$.

- **Intermolecular attraction.** A molecule about to strike the wall is pulled back by the molecules behind it, so the pressure measured at the wall is *less* than that given by the molecular motion. The reduction is proportional to the number of molecules striking the wall per unit area ($\propto$ density $\propto 1/v$) and to the number of molecules pulling each of them back ($\propto 1/v$); hence the correction is a/v^2 , with a a constant for the gas.

Combining the two corrections gives the **van der Waals equation of state**:

$$(p + a/v^2)(v - b) = RT \quad \text{or} \quad p = RT/(v - b) - a/v^2$$

(On a molar basis the same form holds with $\bar{v}$ and R_u .) The term a/v^2 is the *pressure correction* for attraction and b the *volume correction* for molecular size. The equation reduces to $p v = RT$ at low density ($v \gg b$, $a/v^2 \ll p$), and it predicts, at least qualitatively, the whole liquid–vapour behaviour of Figure 4.2 (left): above the critical temperature the isotherms are smooth curves, the critical isotherm has a horizontal point of inflexion, and below T_c the isotherms show the unstable "van der Waals loop" that must be replaced by the horizontal two-phase line (Maxwell's equal-area rule).

4.2 The constants a and b from the critical properties

At the critical point the critical isotherm has a horizontal point of inflexion on the p – v diagram (Figure 4.2, left), so both the first and the second derivatives of p with respect to v at constant T vanish:

$$(\partial p / \partial v)_T = -RT_c / (v_c - b)^2 + 2a / v_c^3 = 0 \quad \text{and} \quad (\partial^2 p / \partial v^2)_T = 2RT_c / (v_c - b)^3 - 6a / v_c^4 = 0.$$

From the first, $RT_c / (v_c - b)^2 = 2a / v_c^3$; from the second, $RT_c / (v_c - b)^3 = 3a / v_c^4$. Dividing the first by the second gives $v_c - b = 2v_c / 3$, i.e. **$v_c = 3b$** . Substituting back, $RT_c = 2a(v_c - b)^2 / v_c^3 = 2a(2b)^2 / (27b^3) = 8a / (27b)$, so **$T_c = 8a / (27Rb)$** . Finally, from the equation of state at the critical point, $p_c = RT_c / (v_c - b) - a / v_c^2 = (8a / 27b) / (2b) - a / (9b^2) = 4a / (27b^2) - 3a / (27b^2) = a / (27b^2)$. Solving for a and b in terms of the two most reliably measured critical properties, p_c and T_c :

$$a = 27R^2 T_c^2 / (64p_c) \quad b = RT_c / (8p_c) \quad Z_c = p_c v_c / (RT_c) = 3/8$$

The predicted critical compressibility factor $3/8 = 0.375$ is higher than the measured values (0.23–0.31), which shows the limitation of a two-constant equation near the critical point. Because a and b are obtained from p_c and T_c alone, the van der Waals constants can be written down for any substance whose critical point is known (Worked Example 4.1).

4.3 The Redlich–Kwong equation

Redlich and Kwong (1949) improved the attraction term by making it temperature-dependent and slightly modifying its volume dependence:

$$p = RT / (v - b) - a / [\sqrt{T} \cdot v(v + b)] \quad a = 0.42748 R^2 T_c^{2.5} / p_c \quad b = 0.08664 RT_c / p_c$$

The constants are again obtained from the critical point by the same inflexion conditions ($Z_c = 1/3$). With only two constants the Redlich–Kwong equation is considerably more accurate than van der Waals' for gases over a wide range of pressure and temperature, and it is widely used in engineering calculations of gas behaviour and in the modern cubic equations (Soave–Redlich–Kwong, Peng–Robinson) derived from it.

4.4 The Beattie–Bridgeman equation

The Beattie–Bridgeman equation (1928) is a five-constant empirical equation, written on a molar basis:

$$p = (R_u T / \bar{v}^2) (1 - c / (\bar{v} T^3)) (\bar{v} + B) - A / \bar{v}^2 \quad \text{with } A = A_0 (1 - a / \bar{v}), \quad B = B_0 (1 - b / \bar{v})$$

The constants A_0 , a , B_0 , b and c are found by fitting experimental data for each gas (for nitrogen: $A_0 = 136.23 \text{ kPa m}^6/\text{kmol}^2$, $a = 0.02617 \text{ m}^3/\text{kmol}$, $B_0 = 0.05046 \text{ m}^3/\text{kmol}$, $b = -0.00691 \text{ m}^3/\text{kmol}$, $c = 4.20 \times 10^4 \text{ m}^3 \text{ K}^3/\text{kmol}$). A is the attraction term (corresponding to van der Waals' a) and B the co-volume term (corresponding to b), both made functions of the molar volume; the term in c corrects the temperature dependence. The equation is accurate to about 0.5 % for densities up to about 0.8 of the critical density, but it fails near the critical point and requires the five constants to be known. The Benedict–Webb–Rubin equation (eight constants) extends it to about 2.5 times the critical density.

4.5 The virial equation of state

The virial equation expresses the compressibility factor as a power series in $1/v$ (or in p):

$$pv = RT(1 + B/v + C/v^2 + D/v^3 + \dots) \quad \text{or} \quad Z = 1 + B'p + C'p^2 + \dots$$

The **virial coefficients** $B(T)$, $C(T)$, ... depend on temperature only; B describes interactions between pairs of molecules, C between triplets, and so on. They can be calculated from statistical mechanics for an assumed intermolecular potential, which gives the equation a theoretical basis. The series is truncated after one or two terms for moderate pressures; the second virial coefficient B is negative at low temperatures (attraction) and passes through zero at the Boyle temperature. Every other equation of state can be expanded into virial form – the van der Waals equation gives $B = b - a/(RT)$.

4.6 Comparison of the equations of state

Equation	Constants	Basis of constants	Accuracy and range	Remarks
Ideal gas $pv = RT$	none (R)	—	Good only at $p_R \ll 1$ or $T_R > 2$	Simplest; no liquefaction
Van der Waals	2 (a , b)	Critical point	Qualitatively correct everywhere; errors of several % at moderate density, poor near the critical point ($Z_c = 0.375$)	Historically first; predicts the critical point and the liquid–vapour loop
Redlich–Kwong	2 (a , b)	Critical point	Much better than van der Waals for gases over wide p , T ; $Z_c = 0.333$	Basis of the modern cubic equations used in process simulation
Beattie–Bridgeman	5	Fitted to data for each gas	About 0.5 % up to $0.8 \rho_c$; not near the critical point	Needs tabulated constants
Benedict–Webb–Rubin	8	Fitted to data	Good to about $2.5 \rho_c$	Used for hydrocarbons and refrigerants
Virial	2–3 coefficients (functions of T)	Theory or data	Excellent at low and moderate density	Theoretical foundation; diverges at high density

KEY POINT All the two-constant equations reduce to $pv = RT$ as $v \rightarrow \infty$, and all can be written as a cubic in v for given p and T – hence "cubic equations of state". Below the critical temperature the cubic has three real roots: the smallest is the liquid volume, the largest the vapour volume and the middle one is unphysical.

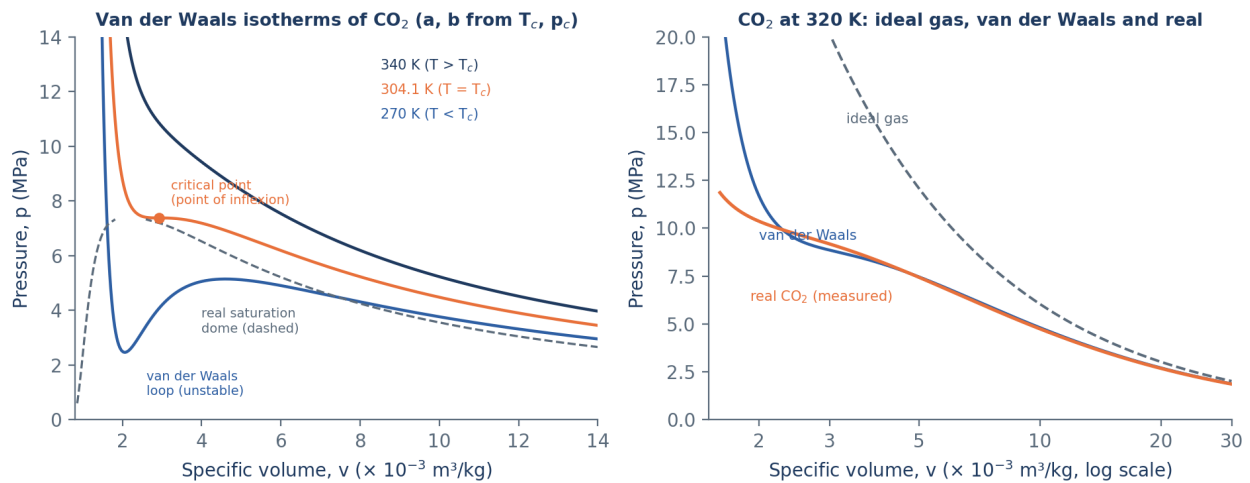


Figure 4.2 (Left) Van der Waals isotherms of carbon dioxide, with the constants a and b obtained from T_c and p_c ; the real saturation dome is shown dashed. (Right) Pressure of carbon dioxide at 320 K against specific volume: ideal-gas, van der Waals and measured values.

4.7 Worked example on the van der Waals equation

WORKED EXAMPLE 4.1 — Van der Waals constants of carbon dioxide and the pressure at a given state

Problem. The critical temperature and pressure of carbon dioxide are 304.2 K and 7.38 MPa, and its molar mass is 44.01 kg/kmol. Determine the van der Waals constants a and b (per kg and per kmol) and the critical specific volume predicted by the equation. Hence estimate the pressure of carbon dioxide at 320 K when its specific volume is 0.004 m³/kg, using (i) the ideal-gas equation and (ii) the van der Waals equation, and compare with the measured value of 8.27 MPa.

Given: $T_c = 304.2 \text{ K}$; $p_c = 7380 \text{ kPa}$; $M = 44.01 \text{ kg/kmol}$; $T = 320 \text{ K}$; $v = 0.004 \text{ m}^3/\text{kg}$; $p_{\text{measured}} = 8.27 \text{ MPa}$.

Formula: $R = R_u/M$; $a = 27R^2T_c^2/(64p_c)$; $b = RT_c/(8p_c)$; $v_c = 3b$; ideal $p = RT/v$; van der Waals $p = RT/(v - b) - a/v^2$.

Solution.

$R = 8.3145/44.01 = 0.1889 \text{ kJ/kg K}$.

$a = 27 \times 0.1889^2 \times 304.2^2/(64 \times 7380) = 27 \times 0.03568 \times 92538/472320 = 0.1888 \text{ kPa m}^6/\text{kg}^2$;

$b = 0.1889 \times 304.2/(8 \times 7380) = 57.46/59040 = 9.73 \times 10^{-4} \text{ m}^3/\text{kg}$.

Per kmol (multiply a by M^2 , b by M): $\bar{a} = 0.1888 \times 44.01^2 = 365.7 \text{ kPa m}^6/\text{kmol}^2$; $\bar{b} = 9.73 \times 10^{-4} \times 44.01 = 0.04284 \text{ m}^3/\text{kmol}$.

Predicted critical volume $v_c = 3b = 2.92 \times 10^{-3} \text{ m}^3/\text{kg}$ (the measured value is $2.14 \times 10^{-3} \text{ m}^3/\text{kg}$ – the van der Waals Z_c of 0.375 is too high).

(i) Ideal gas: $p = 0.1889 \times 320/0.004 = 15114 \text{ kPa} = 15.11 \text{ MPa}$.

(ii) Van der Waals: $RT/(v - b) = 0.1889 \times 320/(0.004 - 0.000973) = 60.45/0.003027 = 19970 \text{ kPa}$;
 $a/v^2 = 0.1888/(0.004)^2 = 11800 \text{ kPa}$; $p = 19970 - 11800 = 8170 \text{ kPa} = 8.17 \text{ MPa}$.

ANSWER $a = 0.1888 \text{ kPa m}^6/\text{kg}^2$ ($365.7 \text{ kPa m}^6/\text{kmol}^2$), $b = 9.73 \times 10^{-4} \text{ m}^3/\text{kg}$ ($0.0428 \text{ m}^3/\text{kmol}$); $p_{\text{ideal}} = 15.1 \text{ MPa}$ (83 % too high), $p_{\text{vdW}} = 8.17 \text{ MPa}$ (1.2 % low) against the measured 8.27 MPa. At this state ($T_R = 1.05$, v less than twice v_c) the attraction term removes more than half of the ideal-gas pressure; the two corrections of van der Waals capture nearly all of the real-gas behaviour (Figure 4.2, right).

5. Reduced properties, the principle of corresponding states and the generalised compressibility chart

5.1 Reduced properties

The **reduced properties** of a substance are its pressure, temperature and specific volume expressed as fractions of the critical values:

$p_R = p/p_c$	$T_R = T/T_c$	$v_R = v/v_c$	$v_R' = v/(RT_c/p_c)$ (pseudo-reduced volume)
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The pseudo-reduced volume v_R' is defined with the *ideal-gas critical volume* RT_c/p_c rather than the measured v_c , because v_c is difficult to measure accurately and because the chart lines are then more nearly universal. Reduced properties are dimensionless and describe how far a state is from the critical point: $p_R \ll 1$ means a low pressure, $T_R \approx 1$ means near the critical temperature.

5.2 The principle of corresponding states

Writing the van der Waals equation in terms of reduced properties eliminates the constants a and b entirely: $(p_R + 3/v_R^2)(3v_R - 1) = 8T_R$. The **reduced equation of state** contains no property of the individual gas, which suggests the **principle (law) of corresponding states**: *all gases, when compared at the same reduced pressure and reduced temperature, have approximately the same compressibility factor, i.e. deviate from ideal-gas behaviour to about the same degree*, $Z = f(p_R, T_R)$, the same function for all substances. Two gases at the same p_R and T_R are said to be in *corresponding states*. The principle is not exact – it holds well for the "simple" nearly spherical molecules (Ar, N₂, O₂, CH₄, CO) and to within a few per cent for most others, less well for strongly polar molecules (H₂O, NH₃) and for the quantum gases H₂, He and Ne, for which corrected critical constants ($T_c + 8$ K, $p_c + 0.8$ MPa) are used – but it is accurate enough for engineering estimates.

5.3 The generalised compressibility chart

Plotting the measured Z of many gases against p_R for lines of constant T_R gives a single set of curves – the **generalised compressibility chart** (Nelson–Obert chart, Figure 4.1, right), which by the principle of corresponding states can be used for any gas whose critical properties are known.

- **Axes and scales.** The horizontal axis is the reduced pressure p_R , usually on a logarithmic scale from about 0.1 to 10 (a low-pressure chart to $p_R = 1$, an intermediate chart to 10 and a high-pressure chart to 40 are published), and the vertical axis is Z on a linear scale from about 0.2 to 1.2 (higher on the high-pressure chart).
- **Constant- T_R curves.** A family of curves for $T_R = 1.0, 1.05, 1.1, 1.2, 1.3, 1.5, 2.0, 3.0 \dots$ Each starts at $Z = 1$ at $p_R \rightarrow 0$, dips below 1 at moderate p_R – the dip being deepest for $T_R = 1$, where Z falls to about 0.25 near $p_R = 1$ – and rises above 1 at high p_R . At T_R above about 2.5 the curves lie close to $Z = 1$ up to $p_R \approx 5$.
- **Pseudo-reduced-volume lines.** A second family of dotted lines of constant v_R' crosses the T_R curves. Since $v_R' = v p_c/(RT_c) = Z T_R/p_R$, each line is a hyperbola-like curve on the chart; the lines allow the chart to be entered when p or T (rather than both) is unknown.
- **Regions of ideal-gas behaviour.** $Z \approx 1$ (within 1–2 %) for $p_R < 0.05$ at any temperature, and for $T_R > 2$ at all pressures except $p_R > 5$. The greatest deviation is near the critical point ($p_R \approx 1$, $T_R \approx 1$) and along the saturation curve.

5.4 Procedure for using the chart

1. Obtain T_c and p_c (and $R = R_u/M$) of the gas from tables.
2. Given p and T : compute p_R and T_R , read Z at their intersection, then $v = ZRT/p$ (Worked Examples 4.2 and 4.3).

3. Given T and v : compute T_R and v_R' , locate the point where the v_R' line crosses the T_R curve, read p_R (and Z), then $p = p_R p_c$ (Worked Example 4.4).
4. Given p and v : compute p_R and v_R' , find the T_R curve through their intersection, then $T = T_R T_c$.
5. Check whether the state lies in an ideal-gas region; if so the chart is unnecessary.

The chart gives Z to about $\pm 2\%$ for simple fluids, which is far better than the ideal-gas equation near the critical region and requires no equation-of-state constants or tables for the particular gas.

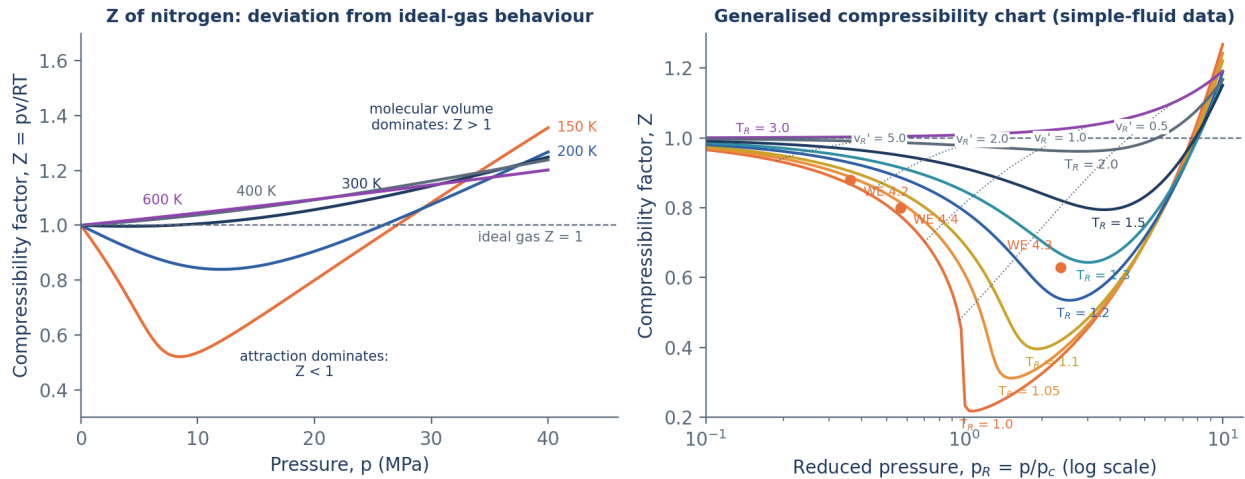


Figure 4.1 (Left) Compressibility factor of nitrogen against pressure at several temperatures. (Right) The generalised compressibility chart constructed from simple-fluid data, with constant- T_R curves, pseudo-reduced-volume lines and the states of Worked Examples 4.2 to 4.4.

5.5 Worked examples on the compressibility chart

WORKED EXAMPLE 4.2 — Specific volume of steam by three methods

Problem. Determine the specific volume of steam at 8 MPa and 400 °C using (i) the ideal-gas equation and (ii) the generalised compressibility chart, and compare with the steam-table value of 0.03432 m³/kg. Find the percentage error of each method and explain, with reference to the reduced properties, why the ideal-gas equation fails while the chart succeeds. Take $R = 0.4615$ kJ/kg K, $T_c = 647.1$ K, $p_c = 22.06$ MPa.

Given: $p = 8000$ kPa; $T = 673.15$ K; $R = 0.4615$ kJ/kg K; $T_c = 647.1$ K; $p_c = 22.06$ MPa; $v_{\text{table}} = 0.03432$ m³/kg.

Formula: $v_{\text{ideal}} = RT/p$; $p_R = p/p_c$, $T_R = T/T_c \rightarrow Z$ from the chart; $v = ZRT/p$; $\text{error} = (v - v_{\text{table}})/v_{\text{table}}$.

Solution.

(i) $v_{\text{ideal}} = 0.4615 \times 673.15/8000 = 0.03883$ m³/kg. Error = $(0.03883 - 0.03432)/0.03432 = +13.1\%$.

(ii) $p_R = 8/22.06 = 0.363$; $T_R = 673.15/647.1 = 1.040$. From the chart (Figure 4.1, right) at $T_R = 1.04$, $p_R = 0.36$: $Z \approx 0.88$.

$v = 0.88 \times 0.03883 = 0.03417$ m³/kg. Error = $(0.03417 - 0.03432)/0.03432 = -0.4\%$.

ANSWER $v_{\text{ideal}} = 0.03883$ m³/kg (13 % high); $v_{\text{chart}} = 0.0342$ m³/kg (0.4 % low); $v_{\text{table}} = 0.03432$ m³/kg. The state lies at $T_R = 1.04$ – only 26 K above the critical temperature – and at more than a third of the critical pressure, close to the saturation dome, where the intermolecular attraction reduces the volume well below the ideal value ($Z = 0.88$). The ideal-gas equation ignores the attraction and over-predicts v by 13 %, an unacceptable error for boiler or turbine design; the chart, through the principle of corresponding states, accounts for it and is within the reading accuracy of the tables.

WORKED EXAMPLE 4.3 — Mass of nitrogen in a high-pressure cylinder

Problem. Nitrogen is stored at 8 MPa and 160 K in a cylinder of volume 0.2 m³. Find the specific volume and the mass of gas using (i) the ideal-gas equation and (ii) the generalised compressibility

chart, and compare with the measured specific volume of $0.003745 \text{ m}^3/\text{kg}$. Take $R = 0.2968 \text{ kJ/kg K}$, $T_c = 126.2 \text{ K}$, $p_c = 3.39 \text{ MPa}$.

Given: $p = 8000 \text{ kPa}$; $T = 160 \text{ K}$; $V = 0.2 \text{ m}^3$; $R = 0.2968 \text{ kJ/kg K}$; $T_c = 126.2 \text{ K}$; $p_c = 3.39 \text{ MPa}$; $v_{\text{measured}} = 0.003745 \text{ m}^3/\text{kg}$.

Formula: $v_{\text{ideal}} = RT/p$; $p_R = p/p_c$, $T_R = T/T_c \rightarrow Z$; $v = Zv_{\text{ideal}}$; $m = V/v$.

Solution.

(i) $v_{\text{ideal}} = 0.2968 \times 160/8000 = 0.005936 \text{ m}^3/\text{kg}$; $m = 0.2/0.005936 = 33.7 \text{ kg}$.

(ii) $p_R = 8/3.39 = 2.36$; $T_R = 160/126.2 = 1.27$. From the chart at $T_R = 1.27$, $p_R = 2.36$: $Z \approx 0.64$.

$v = 0.64 \times 0.005936 = 0.003799 \text{ m}^3/\text{kg}$; $m = 0.2/0.003799 = 52.6 \text{ kg}$.

Measured: $v = 0.003745 \text{ m}^3/\text{kg}$, $m = 0.2/0.003745 = 53.4 \text{ kg}$.

ANSWER (i) Ideal gas: $v = 0.00594 \text{ m}^3/\text{kg}$, $m = 33.7 \text{ kg}$ – the volume is 58 % too high and the mass 37 % too low; (ii) chart: $v = 0.00380 \text{ m}^3/\text{kg}$, $m = 52.6 \text{ kg}$ – within 1.5 % of the measured value. At $p_R = 2.4$ and $T_R = 1.27$ the nitrogen is a dense gas ($Z = 0.63$) far from ideal, and the ideal-gas equation would grossly under-estimate the contents of the cylinder; the chart must be used.

WORKED EXAMPLE 4.4 — Pressure of water vapour by the ideal-gas, van der Waals and chart methods

Problem. Water vapour at 400°C has a specific volume of $0.020 \text{ m}^3/\text{kg}$. Determine its pressure using (i) the ideal-gas equation, (ii) the van der Waals equation with constants from the critical properties and (iii) the generalised compressibility chart using the pseudo-reduced volume. Compare with the steam-table value of 12.52 MPa . Take $R = 0.4615 \text{ kJ/kg K}$, $T_c = 647.1 \text{ K}$, $p_c = 22.06 \text{ MPa}$.

Given: $T = 673.15 \text{ K}$; $v = 0.020 \text{ m}^3/\text{kg}$; $R = 0.4615 \text{ kJ/kg K}$; $T_c = 647.1 \text{ K}$; $p_c = 22060 \text{ kPa}$; $p_{\text{table}} = 12.52 \text{ MPa}$.

Formula: $p_{\text{ideal}} = RT/v$; $a = 27R^2T_c^2/(64p_c)$, $b = RT_c/(8p_c)$, $p_{\text{vdW}} = RT/(v - b) - a/v^2$; $T_R = T/T_c$, $v_R' = v/(RT_c/p_c) \rightarrow p_R$ from the chart, $p = p_R p_c$.

Solution.

(i) $p_{\text{ideal}} = 0.4615 \times 673.15/0.020 = 15533 \text{ kPa} = 15.53 \text{ MPa}$ (error +24 %).

(ii) $a = 27 \times 0.4615^2 \times 647.1^2/(64 \times 22060) = 27 \times 0.21298 \times 418738/1411840 = 1.7055 \text{ kPa m}^6/\text{kg}^2$;

$b = 0.4615 \times 647.1/(8 \times 22060) = 298.64/176480 = 1.692 \times 10^{-3} \text{ m}^3/\text{kg}$.

$RT/(v - b) = 0.4615 \times 673.15/(0.020 - 0.001692) = 310.66/0.018308 = 16969 \text{ kPa}$; $a/v^2 = 1.7055/0.0004 = 4264 \text{ kPa}$.

$p_{\text{vdW}} = 16969 - 4264 = 12705 \text{ kPa} = 12.70 \text{ MPa}$ (error +1.5 %).

(iii) $T_R = 673.15/647.1 = 1.040$; $RT_c/p_c = 0.4615 \times 647.1/22060 = 0.013537 \text{ m}^3/\text{kg}$; $v_R' = 0.020/0.013537 = 1.48$.

Entering the chart at $T_R = 1.04$ along the line $v_R' = 1.48$ gives $p_R \approx 0.56$ ($Z \approx 0.80$); $p = 0.56 \times 22.06 = 12.4 \text{ MPa}$ (error -1 %).

ANSWER $p_{\text{ideal}} = 15.5 \text{ MPa}$ (+24 %), $p_{\text{vdW}} = 12.7 \text{ MPa}$ (+1.5 %), $p_{\text{chart}} = 12.4 \text{ MPa}$ (-1 %), $p_{\text{table}} = 12.52 \text{ MPa}$. At $T_R = 1.04$ and this density ($Z = 0.81$) the attractive forces reduce the pressure by a quarter; the ideal-gas equation is useless, while both the van der Waals equation (constants from the critical point) and the chart give the pressure to within about 1.5 %. For design work the steam tables are used; the chart and cubic equations are the tools when no tables exist.

6. Air and steam compared – constant-volume and constant-pressure processes

Whether a fluid is an ideal gas or a vapour near saturation makes a large difference to the outcome of the same process. Two worked examples compare air (ideal gas) and steam (from the tables) for identical initial conditions and constraints (Figure 4.3).

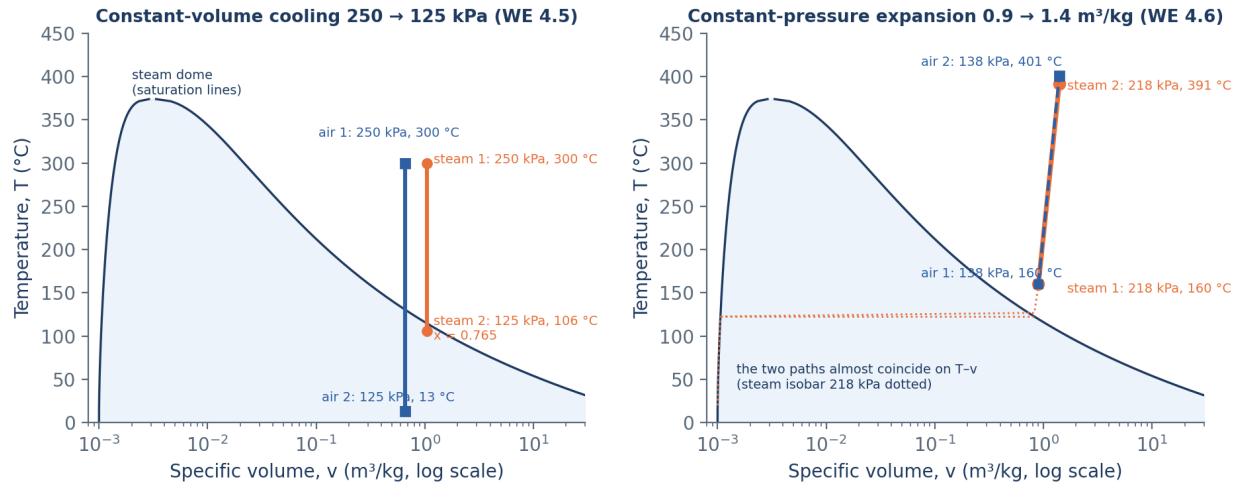


Figure 4.3 Air (blue) and steam (orange) starting from the same state on the T - v diagram: (left) constant-volume cooling from 250 kPa to 125 kPa (Worked Example 4.5); (right) constant-pressure expansion from 0.9 to 1.4 m³/kg (Worked Example 4.6).

WORKED EXAMPLE 4.5 — Constant-volume cooling of air and of steam

Problem. A fluid at 250 kPa and 300 °C occupies a rigid vessel of 0.6 m³. It is cooled at constant volume until the pressure falls to 125 kPa. Find the mass of fluid, the final temperature and the heat transferred if the fluid is (i) air ($R = 0.287$, $c_v = 0.718$ kJ/kg K) and (ii) steam. Show the processes on the T - v plane and explain the difference.

Given: $p_1 = 250$ kPa; $T_1 = 573.15$ K; $V = 0.6$ m³; $p_2 = 125$ kPa. Steam at 250 kPa, 300 °C: $v_1 = 1.0517$ m³/kg, $u_1 = 2807.9$ kJ/kg. Saturation at 125 kPa: $T_{\text{sat}} = 106.0$ °C, $v_f = 0.001048$, $v_g = 1.3749$ m³/kg, $u_f = 444.2$, $u_g = 2513.0$ kJ/kg.

Formula: Air: $m = p_1 V / (RT_1)$; at constant volume $T_2/T_1 = p_2/p_1$; $Q = mc_v(T_2 - T_1)$. Steam: $m = V/v_1$; $v_2 = v_1$; $x_2 = (v_2 - v_f)/(v_g - v_f)$; $u_2 = u_f + x_2(u_g - u_f)$; $Q = m(u_2 - u_1)$.

Solution.

(i) Air: $m = 250 \times 0.6 / (0.287 \times 573.15) = 150/164.5 = 0.912$ kg.

$T_2 = 573.15 \times 125/250 = 286.6$ K = 13.4 °C.

$Q = 0.912 \times 0.718 \times (286.6 - 573.15) = -187.6$ kJ.

(ii) Steam: $m = 0.6/1.0517 = 0.5705$ kg.

At 125 kPa, $v_2 = 1.0517$ m³/kg lies between v_f and v_g , so the steam is wet at $T_2 = 106.0$ °C, with $x_2 = (1.0517 - 0.001048)/(1.3749 - 0.001048) = 1.0507/1.3739 = 0.765$.

$u_2 = 444.2 + 0.765 \times (2513.0 - 444.2) = 444.2 + 1582.6 = 2026.8$ kJ/kg.

$Q = 0.5705 \times (2026.8 - 2807.9) = -445.6$ kJ.

ANSWER Air: $m = 0.912$ kg, $T_2 = 13.4$ °C, $Q = -187.6$ kJ. Steam: $m = 0.571$ kg, $T_2 = 106.0$ °C, $x_2 = 0.765$, $Q = -446$ kJ. Although the vessel holds only 63 % as much steam as air (steam's R is larger, so its specific volume is larger), the steam gives up 2.4 times as much heat: halving the pressure of the air simply halves its absolute temperature (a fall of 287 K, sensible heat only), whereas the steam reaches the saturated vapour line at 114.5 °C (166 kPa), after which the pressure falls by condensation with only a small further fall of temperature, and most of the heat removed is latent heat (23.5 % of the steam condenses). On the T - v plane both processes are vertical lines; the air

line runs from 300 °C to 13 °C in the superheated region, the steam line crosses the saturated vapour line and ends inside the dome (Figure 4.3, left).

WORKED EXAMPLE 4.6 — Constant-pressure expansion of air and of steam

Problem. One kilogram of fluid at 160 °C with a specific volume of 0.90 m³/kg expands at constant pressure until its specific volume is 1.40 m³/kg. Find the pressure, the work done, the heat transferred and the final temperature if the fluid is (i) air ($R = 0.287$, $c_p = 1.005$ kJ/kg K) and (ii) steam, and explain why the steam absorbs much more heat than the air.

Given: $T_1 = 433.15$ K; $v_1 = 0.90$ m³/kg; $v_2 = 1.40$ m³/kg; $m = 1$ kg. Steam tables: at 160 °C, $v_g = 0.3068$ m³/kg (so steam with $v = 0.90$ is superheated); superheated steam with $v = 0.90$ m³/kg at 160 °C has $p = 218.4$ kPa, $h_1 = 2788.4$ kJ/kg; at 218.4 kPa and $v = 1.40$ m³/kg the temperature is 391.4 °C and $h_2 = 3258.8$ kJ/kg.

Formula: Air: $p = RT_1/v_1$; $W = p(v_2 - v_1)$; $T_2 = T_1 v_2/v_1$; $Q = c_p(T_2 - T_1)$. Steam: p and h_1 from the tables at (160 °C, v_1); $W = p(v_2 - v_1)$; T_2 and h_2 from the tables at (p , v_2); $Q = h_2 - h_1$.

Solution.

(i) Air: $p = 0.287 \times 433.15/0.90 = 138.1$ kPa.

$W = 138.1 \times (1.40 - 0.90) = 69.1$ kJ.

$T_2 = 433.15 \times 1.40/0.90 = 673.8$ K = 400.6 °C.

$Q = 1.005 \times (673.8 - 433.15) = 241.8$ kJ ($\Delta U = 0.718 \times 240.6 = 172.8$ kJ; $Q = \Delta U + W = 172.8 + 69.1 = 241.9$ kJ ✓).

(ii) Steam: $p = 218.4$ kPa (superheated, 160 °C; $T_{\text{sat}} = 123.0$ °C).

$W = 218.4 \times (1.40 - 0.90) = 109.2$ kJ.

$T_2 = 391.4$ °C (by interpolation in the superheat table at 218.4 kPa for $v = 1.40$ m³/kg).

$Q = h_2 - h_1 = 3258.8 - 2788.4 = 470.4$ kJ.

ANSWER Air: $p = 138.1$ kPa, $W = 69.1$ kJ, $Q = 241.8$ kJ, $T_2 = 400.6$ °C. Steam: $p = 218.4$ kPa, $W = 109.2$ kJ, $Q = 470.4$ kJ, $T_2 = 391.4$ °C. For the same initial temperature and volume the steam is at a 58 % higher pressure (its gas constant is 0.4615 against 0.287 kJ/kg K), so it does 58 % more work; and its specific heat at constant pressure (about 2.0 kJ/kg K, twice that of air) means that raising it through nearly the same 230 K needs about twice the heat. Both effects together make Q for steam almost double that for air. The final temperatures are close because the steam, being well superheated, behaves nearly as an ideal gas ($Z = 0.98$ at state 1); the small difference comes from the deviation of Z from unity.

7. Thermodynamic relations – partial derivatives, thermodynamic potentials and the Maxwell relations

Of the thermodynamic properties only p , v and T (and, with more effort, the specific heats, the expansivity and the compressibility) can be measured directly; u , h , s , f and g cannot. The **general thermodynamic relations** express the derivatives of the unmeasurable properties in terms of the measurable ones, so that tables such as the steam tables can be constructed from p – v – T data and a few specific-heat measurements. They rest on the fact that every property is a point function, so that its differential is exact.

7.1 Mathematical preliminaries

Let $z = z(x, y)$ be a property expressed in terms of two other properties. Then

$$dz = (\partial z / \partial x)_y dx + (\partial z / \partial y)_x dy = M dx + N dy,$$

and because the order of differentiation is immaterial, the **condition for an exact differential** is

$$(\partial M/\partial y)_x = (\partial N/\partial x)_y \quad [z \text{ is a property} \Leftrightarrow dz \text{ is exact}]$$

Two further relations follow from the same equation. Setting $dz = 0$ gives the **cyclic relation** $(\partial x/\partial y)_z(\partial y/\partial z)_x(\partial z/\partial x)_y = -1$, and holding one variable constant gives the **reciprocity relation** $(\partial x/\partial y)_z = 1/(\partial y/\partial x)_z$. Three measurable derivatives of v recur throughout: the **coefficient of volume expansion** $\beta = (1/v)(\partial v/\partial T)_p$, the **isothermal compressibility** $\kappa_T = -(1/v)(\partial v/\partial p)_T$ and the **isentropic compressibility** $\kappa_s = -(1/v)(\partial v/\partial p)_s$.

7.2 The thermodynamic potentials

Four energy-like properties, the **thermodynamic potentials**, and their exact differentials (from the first Tds equation $du = T ds - p dv$ and the definitions) are (Figure 4.4, left):

Potential	Definition	Differential	Natural variables
Internal energy u	—	$du = T ds - p dv$	(s, v)
Enthalpy h	$h = u + pv$	$dh = T ds + v dp$	(s, p)
Helmholtz function f	$f = u - Ts$	$df = -s dT - p dv$	(T, v)
Gibbs function g	$g = h - Ts$	$dg = -s dT + v dp$	(T, p)

The **Helmholtz function** $f = u - Ts$ is the maximum work obtainable from a closed system in an isothermal process (it is the "free energy" available at constant T and v); the **Gibbs function** $g = h - Ts$ plays the same role for processes at constant T and p and governs phase and chemical equilibrium (the Gibbs functions of two phases in equilibrium are equal, $g_f = g_g$, which is the basis of the Clapeyron equation). Each potential, expressed in its natural variables, contains all the thermodynamic information about the substance: from $u(s, v)$, for example, $T = (\partial u/\partial s)_v$ and $p = -(\partial u/\partial v)_s$.

7.3 Derivation of the Maxwell relations

Each of the four differentials is exact, so the condition $(\partial M/\partial y)_x = (\partial N/\partial x)_y$ applies to each. From $du = T ds - p dv$, with $M = T$, $N = -p$, $x = s$, $y = v$: $(\partial T/\partial v)_s = -(\partial p/\partial s)_v$. From $dh = T ds + v dp$: $(\partial T/\partial p)_s = (\partial v/\partial s)_p$. From $df = -s dT - p dv$: $(\partial s/\partial v)_T = (\partial p/\partial T)_v$. From $dg = -s dT + v dp$: $-(\partial s/\partial p)_T = (\partial v/\partial T)_p$. These are the four **Maxwell relations**:

$$(\partial T/\partial v)_s = -(\partial p/\partial s)_v \quad (\partial T/\partial p)_s = (\partial v/\partial s)_p \quad (\partial p/\partial T)_v = (\partial s/\partial v)_T \quad (\partial v/\partial T)_p = -(\partial s/\partial p)_T$$

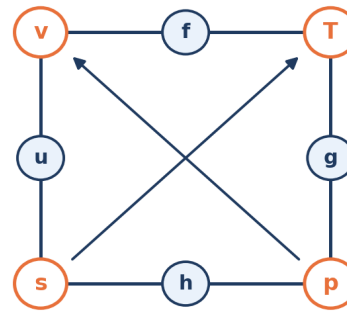
Their value lies in the last two: the change of entropy with volume or pressure at constant temperature – which cannot be measured – equals a p – v – T derivative that can. The relations are remembered with the **thermodynamic square** (Figure 4.4, right): the corners carry v , T , p , s (clockwise from top left), the sides carry the potential whose natural variables are the adjacent corners (f between v and T , g between T and p , h between p and s , u between s and v), and the diagonal arrows give the signs.

The four thermodynamic potentials

Internal energy u	du = T ds - p dv natural variables: u = u(s, v)
Enthalpy h = u + pv	dh = T ds + v dp natural variables: h = h(s, p)
Helmholtz function f = u - Ts	df = -s dT - p dv natural variables: f = f(T, v)
Gibbs function g = h - Ts	dg = -s dT + v dp natural variables: g = g(T, p)

Thermodynamic (Born) square - mnemonic for the Maxwell relations

"Very Tall People Sometimes Understand Hard Formulas Greatly":
v, T, p, s round the corners, u, h, f, g on the sides



Each potential sits between its natural variables; the arrows give the signs.
Maxwell: $(\partial T/\partial v)_s = -(\partial p/\partial s)_v$ $(\partial T/\partial p)_s = (\partial v/\partial s)_p$
 $(\partial p/\partial T)_v = (\partial s/\partial v)_T$ $(\partial v/\partial T)_p = -(\partial s/\partial p)_T$

Figure 4.4 (Left) The four thermodynamic potentials with their differentials and natural variables. (Right) The thermodynamic (Born) square used as a mnemonic for the Maxwell relations.

8. The Tds equations, the difference and ratio of specific heats and the energy equation

8.1 The first Tds equation

Take $s = s(T, v)$: $ds = (\partial s/\partial T)_v dT + (\partial s/\partial v)_T dv$. At constant volume the first law gives $T ds = du = c_v dT$, so $(\partial s/\partial T)_v = c_v/T$; and by the third Maxwell relation $(\partial s/\partial v)_T = (\partial p/\partial T)_v$. Hence

$$T ds = c_v dT + T(\partial p/\partial T)_v dv \quad (\text{first Tds equation})$$

8.2 The second Tds equation

Take $s = s(T, p)$: $ds = (\partial s/\partial T)_p dT + (\partial s/\partial p)_T dp$. At constant pressure $T ds = dh = c_p dT$, so $(\partial s/\partial T)_p = c_p/T$; and by the fourth Maxwell relation $(\partial s/\partial p)_T = -(\partial v/\partial T)_p$. Hence

$$T ds = c_p dT - T(\partial v/\partial T)_p dp \quad (\text{second Tds equation})$$

Both equations express the entropy change in terms of measurable quantities only – c_v or c_p and the p – v – T behaviour. (The equations $T ds = du + p dv$ and $T ds = dh - v dp$ of Unit II are the same equations before the derivatives of u and h are eliminated.)

Application to an ideal gas. With $p v = RT$, $(\partial p/\partial T)_v = R/v$ and $(\partial v/\partial T)_p = R/p$, so the two equations give $ds = c_v dT/T + R dv/v$ and $ds = c_p dT/T - R dp/p$, which integrate to

$$s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln(v_2/v_1) = c_p \ln(T_2/T_1) - R \ln(p_2/p_1)$$

the results used in Unit II. For a van der Waals gas $(\partial p/\partial T)_v = R/(v - b)$, so $s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln[(v_2 - b)/(v_1 - b)]$; for a liquid or solid the second equation with $v(\partial v/\partial T)_p$ small gives $ds \approx c_p dT/T$.

8.3 The difference of the specific heats

Equating the two Tds equations, $c_p dT - T(\partial v/\partial T)_p dp = c_v dT + T(\partial p/\partial T)_v dv$, and solving for dT :

$$dT = [T(\partial p/\partial T)_v/(c_p - c_v)] dv + [T(\partial v/\partial T)_p/(c_p - c_v)] dp.$$

But $T = T(v, p)$ also gives $dT = (\partial T/\partial v)_p dv + (\partial T/\partial p)_v dp$; comparing the coefficients of dv (or of dp – both give the same result),

$$c_p - c_v = T(\partial p/\partial T)_v(\partial v/\partial T)_p.$$

By the cyclic relation $(\partial p/\partial T)_v(\partial T/\partial v)_p(\partial v/\partial p)_T = -1$, i.e. $(\partial p/\partial T)_v = -(\partial v/\partial T)_p(\partial p/\partial v)_T$, so

$$c_p - c_v = -T(\partial v/\partial T)_p^2(\partial p/\partial v)_T = T v \beta^2/\kappa_T$$

using $\beta = (1/v)(\partial v/\partial T)_p$ and $\kappa_T = -(1/v)(\partial v/\partial p)_T$. Three important conclusions follow. (i) For an **ideal gas** $(\partial v/\partial T)_p = R/p$ and $(\partial p/\partial v)_T = -RT/v^2$, so $c_p - c_v = -T(R/p)^2(-RT/v^2) = R^2T^2/(p^2v^2) \cdot R = R$ (since $pv = RT$) – the familiar result. (ii) Since $(\partial p/\partial v)_T$ is negative for every known substance (compressing a substance isothermally always raises its pressure – the condition of mechanical stability) and $(\partial v/\partial T)_p^2$ is never negative, **$c_p - c_v \geq 0$: c_p can never be less than c_v** ; they are equal when $(\partial v/\partial T)_p = 0$, as for water at 4 °C where its density is a maximum. (iii) As **$T \rightarrow 0$, $c_p - c_v \rightarrow 0$** : the two specific heats become equal at absolute zero. For liquids and solids the difference is small because β is small (Worked Example 4.7), and the single symbol c is used.

8.4 The ratio of the specific heats

Set $ds = 0$ in the two Tds equations to describe an isentropic process: from the first, $c_v dT = -T(\partial p/\partial T)_v dv$, i.e. $(\partial T/\partial v)_s = -(T/c_v)(\partial p/\partial T)_v$; from the second, $c_p dT = T(\partial v/\partial T)_p dp$, i.e. $(\partial T/\partial p)_s = (T/c_p)(\partial v/\partial T)_p$. Dividing the first by the second,

$$(\partial p/\partial v)_s = (\partial T/\partial v)_s/(\partial T/\partial p)_s = -(c_p/c_v)(\partial p/\partial T)_v/(\partial v/\partial T)_p = (c_p/c_v)(\partial p/\partial v)_T$$

(using the cyclic relation once more). Therefore

$$\gamma = c_p/c_v = (\partial p/\partial v)_s/(\partial p/\partial v)_T = \kappa_T/\kappa_s \quad (\text{slope of isentrope} = \gamma \times \text{slope of isotherm on the } p\text{--}v \text{ plane})$$

Since $\gamma > 1$, the isentropic curve through any state is steeper than the isotherm through the same state on the p – v diagram – the reason why an adiabatic compression raises the pressure more than an isothermal one, and why the isentropic compressibility κ_s is smaller than the isothermal κ_T (the velocity of sound, $c^2 = 1/(\rho\kappa_s) = \gamma/(\rho\kappa_T)$, is $\gamma^{1/2}$ times what it would be for isothermal compression). For an ideal gas $(\partial p/\partial v)_T = -p/v$, so $(\partial p/\partial v)_s = -\gamma p/v$, which integrates to $pv^\gamma = \text{constant}$.

8.5 The energy equation

Substituting the first Tds equation into $du = T ds - p dv$ gives $du = c_v dT + [T(\partial p/\partial T)_v - p] dv$, and comparing with $du = (\partial u/\partial T)_v dT + (\partial u/\partial v)_T dv$:

$$(\partial u/\partial v)_T = T(\partial p/\partial T)_v - p \quad (\text{energy equation relation}) \quad (\partial h/\partial p)_T = v - T(\partial v/\partial T)_p \quad (\text{enthalpy relation})$$

the second following in the same way from $dh = T ds + v dp$ with the second Tds equation. The energy equation gives the change of internal energy with volume at constant temperature – the effect of the intermolecular forces – in terms of the equation of state. **Ideal gas:** $(\partial p/\partial T)_v = R/v$, so $(\partial u/\partial v)_T = RT/v - p = 0$; and $(\partial v/\partial T)_p = R/p$, so $(\partial h/\partial p)_T = v - RT/p = 0$. *The internal energy and the enthalpy of an ideal gas are functions of temperature alone* – Joule's law is thus a consequence of $pv = RT$ and the second law. Differentiating the energy equation again, $(\partial c_v/\partial v)_T = T(\partial^2 p/\partial T^2)_v$, which is zero for an ideal gas (and for a van der Waals gas): c_v then depends on T only. **Van der Waals gas:** $(\partial p/\partial T)_v = R/(v - b)$, so $(\partial u/\partial v)_T = RT/(v - b) - p = a/v^2$: the internal energy increases with volume at constant temperature, because work must be done against the attractive forces to separate the molecules – which is why a real gas cools in a free expansion (Joule) and, usually, on throttling (Joule–Thomson).

WORKED EXAMPLE 4.7 — Difference of the specific heats for liquid water and for copper

Problem. For liquid water at 20 °C and 1 atm the coefficient of volume expansion is $\beta = 2.07 \times 10^{-4} \text{ K}^{-1}$, the isothermal compressibility $\kappa_T = 4.59 \times 10^{-10} \text{ Pa}^{-1}$ and the specific volume 1.002×10^{-3}

m³/kg; for copper at 300 K, $\beta = 5.0 \times 10^{-5} \text{ K}^{-1}$, $\kappa_T = 7.3 \times 10^{-12} \text{ Pa}^{-1}$ and $v = 1.116 \times 10^{-4} \text{ m}^3/\text{kg}$. Calculate $c_p - c_v$ for each and comment, given that $c_p = 4.184 \text{ kJ/kg K}$ for water and 0.385 kJ/kg K for copper.

Given: as stated; $T = 293.15 \text{ K}$ (water), 300 K (copper).

Formula: $c_p - c_v = T v \beta^2 / \kappa_T$ (J/kg K when v in m³/kg, κ_T in Pa⁻¹).

Solution.

Water: $c_p - c_v = 293.15 \times 1.002 \times 10^{-3} \times (2.07 \times 10^{-4})^2 / (4.59 \times 10^{-10}) = 293.15 \times 1.002 \times 10^{-3} \times 4.285 \times 10^{-8} / (4.59 \times 10^{-10}) = 27.4 \text{ J/kg K} = 0.0274 \text{ kJ/kg K}$.

Hence $c_v = 4.184 - 0.027 = 4.157 \text{ kJ/kg K}$ and $\gamma = 4.184 / 4.157 = 1.0066$.

Copper: $c_p - c_v = 300 \times 1.116 \times 10^{-4} \times (5.0 \times 10^{-5})^2 / (7.3 \times 10^{-12}) = 300 \times 1.116 \times 10^{-4} \times 2.5 \times 10^{-9} / (7.3 \times 10^{-12}) = 11.5 \text{ J/kg K} = 0.0115 \text{ kJ/kg K}$; $c_v = 0.385 - 0.012 = 0.373 \text{ kJ/kg K}$, $\gamma = 1.031$.

ANSWER $c_p - c_v = 0.027 \text{ kJ/kg K}$ for water (0.65 % of c_p) and 0.012 kJ/kg K for copper (3 % of c_p). For liquids and solids the two specific heats are nearly equal – hence the single value c quoted in tables – whereas for an ideal gas the difference is R (0.287 kJ/kg K for air, 29 % of c_p). The difference is a real, measurable consequence of thermal expansion working against the compressibility.

9. The Joule–Thomson coefficient and the inversion curve

9.1 The porous-plug experiment

Joule and Thomson (Kelvin) forced a gas steadily through a porous plug (a throttle) in an insulated pipe and measured the temperatures on the two sides (Figure 4.5, left). With no heat transfer, no work and negligible change of velocity the steady flow energy equation gives $h_2 = h_1$: throttling is a constant-enthalpy process. For an ideal gas h depends on T alone, so the temperature would be unchanged; for real gases a temperature change is observed – most gases cool at room temperature, but hydrogen and helium warm. The **Joule–Thomson (Joule–Kelvin) coefficient** is the slope of a constant-enthalpy line on the T – p plane:

$\mu_J = (\partial T / \partial p)_h$ $\mu_J > 0$: temperature falls on throttling (cooling); $\mu_J < 0$: temperature rises;
 $\mu_J = 0$: inversion

Repeating the experiment from a fixed inlet state with different exit pressures traces a constant-enthalpy (isenthalpic) curve through the measured (p_2 , T_2) points; repeating it from other inlet states gives a family of isenthalps (Figure 4.5, right). Each isenthalp has a maximum – the point where $\mu_J = 0$ – and the locus of these maxima is the **inversion curve**. Inside the inversion curve (lower pressures and temperatures) $\mu_J > 0$ and throttling cools the gas; outside it $\mu_J < 0$ and throttling warms the gas. The inversion curve meets the $p = 0$ axis at the **maximum inversion temperature**: above it a gas cannot be cooled by throttling at any pressure (nitrogen 621 K, air $\approx$ 603 K, hydrogen 202 K, helium 40 K – which is why hydrogen and helium must be pre-cooled before they can be liquefied by throttling).

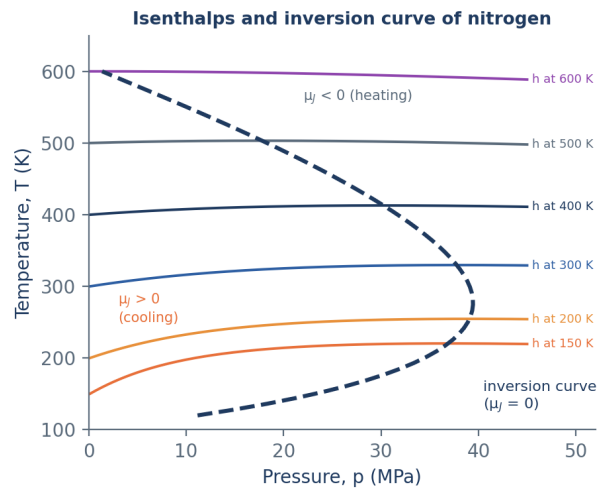
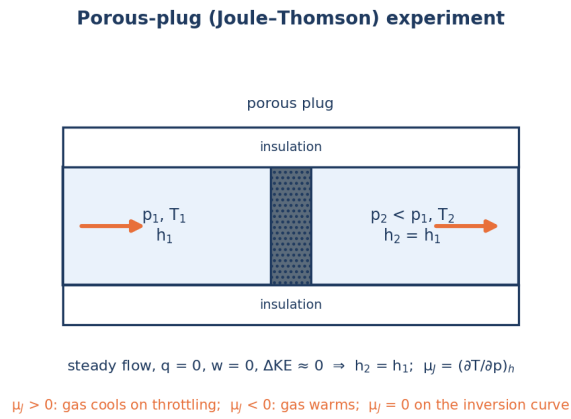


Figure 4.5 (Left) The porous-plug experiment. (Right) Constant-enthalpy curves and the inversion curve of nitrogen on the T - p plane; inside the inversion curve throttling cools the gas.

9.2 Derivation of the expression for the Joule–Thomson coefficient

Write $h = h(T, p)$: $dh = c_p dT + (\partial h / \partial p)_T dp$, and use the enthalpy relation of Section 8.5, $(\partial h / \partial p)_T = v - T(\partial v / \partial T)_p$:

$$dh = c_p dT + [v - T(\partial v / \partial T)_p] dp.$$

For a throttling process $dh = 0$, so

$$\mu_J = (\partial T / \partial p)_h = (1/c_p)[T(\partial v / \partial T)_p - v] = (v/c_p)(T\beta - 1)$$

Ideal gas: $(\partial v / \partial T)_p = R/p$, so $T(\partial v / \partial T)_p = RT/p = v$ and $\mu_J = 0$ – an ideal gas suffers no temperature change on throttling, consistent with $h = h(T)$. **Real gas:** the sign of μ_J is the sign of $[T(\partial v / \partial T)_p - v]$, i.e. of $(T\beta - 1)$. At low pressure and moderate temperature the attractive forces make v increase with T faster than in an ideal gas ($T\beta > 1$) and $\mu_J > 0$; at high pressure or high temperature the finite molecular volume dominates ($T\beta < 1$) and $\mu_J < 0$. For a van der Waals gas at low pressure the expression reduces to $\mu_J \approx (1/c_p)(2a/(RT) - b)$, so the gas cools when $T < 2a/(Rb)$ and the (low-pressure) **inversion temperature is $T_i = 2a/(Rb) = 27T_c/4$** – about 6.75 times the critical temperature; measured maximum inversion temperatures are lower (about $5 T_c$), the van der Waals value being only a rough guide. The Joule–Thomson effect is the basis of the Linde–Hampson process for liquefying gases and of the expansion valve in vapour-compression refrigerators, and μ_J measurements are used to determine the enthalpy of real gases.

WORKED EXAMPLE 4.8 — Joule–Thomson coefficient of steam and the inversion temperature of nitrogen

Problem. (i) Steam at 2 MPa and 300 °C is throttled to 1 MPa. Using the steam tables, find the exit temperature and estimate the mean Joule–Thomson coefficient of steam over this range. (ii) Estimate the low-pressure inversion temperature and the Boyle temperature of nitrogen from the van der Waals constants ($T_c = 126.2$ K) and compare with the measured values of 621 K and 327 K.

Given: (i) At 2 MPa, 300 °C: $h = 3024.3$ kJ/kg; superheat table at 1 MPa: 250 °C $h = 2943.1$, 300 °C $h = 3051.6$ kJ/kg. (ii) Van der Waals: $T_i = 2a/(Rb) = 27T_c/4$; $T_B = a/(Rb) = 27T_c/8$.

Formula: Throttling $h_2 = h_1$; $\mu_J \approx \Delta T / \Delta p$ along the isenthalp; $T_i = 6.75 T_c$, $T_B = 3.375 T_c$.

Solution.

(i) At 1 MPa, $h = 3024.3$ lies between 2943.1 (250 °C) and 3051.6 (300 °C): $T_2 = 250 + 50 \times (3024.3 - 2943.1) / (3051.6 - 2943.1) = 250 + 50 \times 0.748 = 287.4$ °C.

$\mu_J \approx (T_2 - T_1)/(p_2 - p_1) = (287.4 - 300)/(1 - 2) = 12.6 \text{ K/MPa}$ (positive: the steam cools by 12.6 K).
(ii) $T_i = 6.75 \times 126.2 = 852 \text{ K}$; $T_B = 3.375 \times 126.2 = 426 \text{ K}$.

ANSWER (i) $T_2 \approx 287^\circ\text{C}$, $\mu_J \approx 12.6 \text{ K/MPa}$ – steam near its saturation region has a large positive Joule–Thomson coefficient (an ideal gas would show no change); (ii) the van der Waals estimates 852 K and 426 K are 37 % and 30 % above the measured 621 K and 327 K – the two-constant equation gives the right order of magnitude and the correct 2:1 ratio T_i/T_B , but not accurate values.

10. The Clapeyron and Clausius–Clapeyron equations and phase-change processes

10.1 The Clapeyron equation

During a phase change (melting, vaporisation, sublimation) at constant pressure the temperature is also constant, and the saturation pressure is a function of temperature alone: $p = p_{\text{sat}}(T)$. Apply the third Maxwell relation, $(\partial p/\partial T)_v = (\partial s/\partial v)_T$, to the two-phase region. Since p depends only on T , the left side is the slope of the saturation curve on the p – T diagram, $(dp/dT)_{\text{sat}}$, which is independent of v ; the right side, at constant T (and hence constant p), is the ratio of the entropy change to the volume change during the phase change, s_{fg}/v_{fg} (Figure 4.6, left). Hence

$$(dp/dT)_{\text{sat}} = s_{fg}/v_{fg} = (s_g - s_f)/(v_g - v_f).$$

Because the phase change is a reversible isothermal process, $T s_{fg} = h_{fg}$ (Unit III), and so

$$(dp/dT)_{\text{sat}} = h_{fg}/(T v_{fg}) \quad (\text{Clapeyron equation})$$

The equation is exact and applies to any first-order phase change: for fusion $(dp/dT)_{\text{fus}} = h_{sf}/(T v_{sf})$ and for sublimation $(dp/dT)_{\text{sub}} = h_{ig}/(T v_{ig})$. It links a measurable slope of the phase boundary with the latent heat and the volume change, so that the latent heat can be found from p – v – T data alone (Worked Example 4.9). (The same result follows from the equality of the Gibbs functions of the two phases, $g_f = g_g$, along the saturation line: $dg_f = dg_g$ gives $-s_f dT + v_f dp = -s_g dT + v_g dp$.)

10.2 The Clausius–Clapeyron equation

For the liquid–vapour (or solid–vapour) transition at pressures well below the critical pressure two simplifications are justified: the volume of the liquid is negligible compared with that of the vapour, $v_{fg} \approx v_g$, and the vapour behaves as an ideal gas, $v_g = RT/p$. Then $(dp/dT)_{\text{sat}} = h_{fg} p/(RT^2)$, or

$$(1/p)(dp/dT) = d(\ln p)/dT = h_{fg}/(RT^2) \quad \Rightarrow \quad \ln(p_2/p_1) = (h_{fg}/R)(1/T_1 - 1/T_2) \quad (h_{fg} \text{ constant})$$

(Clausius–Clapeyron equation)

The integrated form assumes, in addition, that h_{fg} is constant over the temperature range considered. It shows that $\ln p_{\text{sat}}$ plotted against $1/T$ is nearly a straight line of slope $-h_{fg}/R$ (Figure 4.6, right), and it is used to estimate the saturation pressure at one temperature from that at another (Worked Example 4.10), to find the latent heat from two vapour-pressure measurements, and to extrapolate vapour-pressure data. Its parameters are p_{sat} , T , h_{fg} and R .

10.3 Phase-change processes

- **Vaporisation.** $v_{fg} > 0$ and $h_{fg} > 0$, so $dp/dT > 0$: the boiling point rises with pressure (pressure cookers; boilers), as the vapour-pressure curve shows.
- **Fusion.** $h_{sf} > 0$ always, but $v_{sf} = v_{\text{liquid}} - v_{\text{solid}}$ is positive for substances that expand on melting (most) and negative for those that contract on melting, notably water (ice floats). For water $(dp/dT)_{\text{fus}}$ is

therefore *negative*: the melting point of ice falls with increasing pressure – by only about 0.0074 K per bar (Worked Example 4.11) – which is why the fusion curve of Figure 3.3 leans backward and why ice melts under a skate blade.

- **Sublimation.** Below the triple point the solid passes directly to vapour; since $h_{ig} = h_{if} + h_{fg}$ (the enthalpy of sublimation is the sum of the enthalpies of fusion and vaporisation), the sublimation curve is steeper than the vaporisation curve at the triple point (Practice Problem P8).

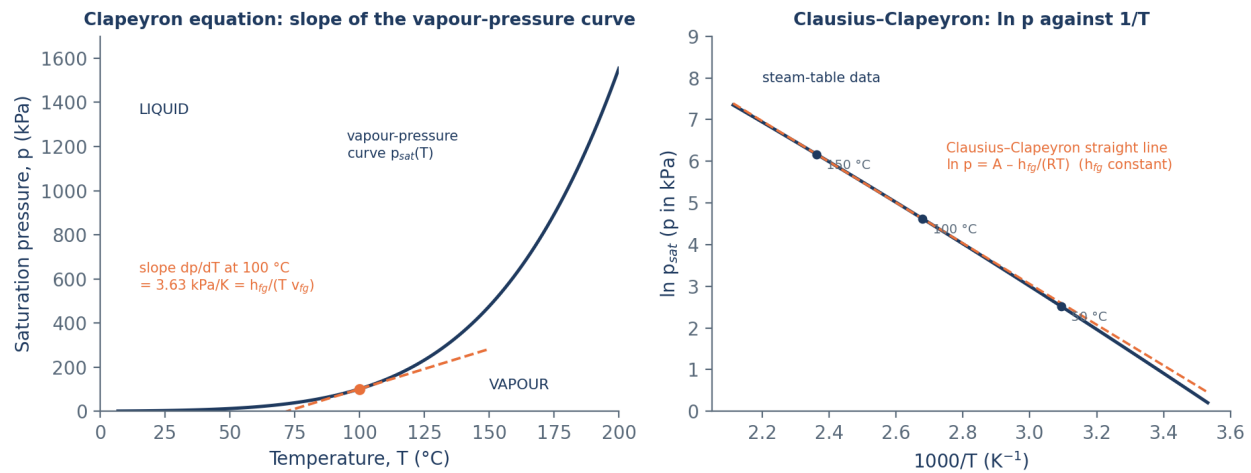


Figure 4.6 (Left) The Clapeyron equation gives the slope of the vapour-pressure curve at any temperature in terms of h_{fg} and v_{fg} . (Right) $\ln p_{sat}$ against $1/T$ for water: the steam-table data and the Clausius–Clapeyron straight line drawn with the latent heat at 100 °C.

WORKED EXAMPLE 4.9 — Latent heat of water from the slope of the vapour-pressure curve

Problem. The saturation pressures of water at 95 °C and 105 °C are 84.61 kPa and 120.90 kPa, and at 100 °C $v_f = 0.001043 \text{ m}^3/\text{kg}$ and $v_g = 1.6720 \text{ m}^3/\text{kg}$. Estimate the enthalpy of vaporisation at 100 °C from the Clapeyron equation and compare with the steam-table value of 2256.5 kJ/kg.

Given: $p(95 \text{ °C}) = 84.61 \text{ kPa}$; $p(105 \text{ °C}) = 120.90 \text{ kPa}$; $T = 373.15 \text{ K}$; $v_{fg} = 1.6720 - 0.001043 = 1.6710 \text{ m}^3/\text{kg}$.

Formula: $(dp/dT)_{sat} \approx \Delta p / \Delta T$ (central difference); $h_{fg} = T v_{fg} (dp/dT)_{sat}$.

Solution.

$(dp/dT)_{sat} \approx (120.90 - 84.61) / (105 - 95) = 36.29 / 10 = 3.629 \text{ kPa/K}$.

$h_{fg} = 373.15 \times 1.6710 \times 3.629 = 2262.8 \text{ kJ/kg}$.

ANSWER $h_{fg} \approx 2263 \text{ kJ/kg}$, within 0.3 % of the table value 2256.5 kJ/kg; the small error comes from replacing the derivative by a finite difference over 10 K. The latent heat has been obtained entirely from p - v - T measurements – the purpose of the thermodynamic relations.

WORKED EXAMPLE 4.10 — Saturation pressure by the Clausius–Clapeyron equation

Problem. Water boils at 100 °C at 101.42 kPa, and its enthalpy of vaporisation is 2256.5 kJ/kg. Estimate the saturation pressure at 110 °C using the Clausius–Clapeyron equation ($R = 0.4615 \text{ kJ/kg K}$) and compare with the table value of 143.38 kPa. State the assumptions.

Given: $p_1 = 101.42 \text{ kPa}$ at $T_1 = 373.15 \text{ K}$; $T_2 = 383.15 \text{ K}$; $h_{fg} = 2256.5 \text{ kJ/kg}$; $R = 0.4615 \text{ kJ/kg K}$.

Formula: $\ln(p_2/p_1) = (h_{fg}/R)(1/T_1 - 1/T_2)$.

Solution.

$h_{fg}/R = 2256.5 / 0.4615 = 4889.5 \text{ K}$; $1/T_1 - 1/T_2 = 1/373.15 - 1/383.15 = 0.0026799 - 0.0026099 = 6.995 \times 10^{-5} \text{ K}^{-1}$.

$\ln(p_2/p_1) = 4889.5 \times 6.995 \times 10^{-5} = 0.3420$; $p_2/p_1 = e^{0.3420} = 1.4078$.
 $p_2 = 1.4078 \times 101.42 = 142.8 \text{ kPa}$.

ANSWER $p_{\text{sat}}(110^\circ\text{C}) \approx 142.8 \text{ kPa}$ against 143.38 kPa from the tables (-0.4%). Assumptions: v_f negligible compared with v_g , the vapour an ideal gas, and h_{fg} constant between 100 and 110°C (it actually falls from 2256.5 to 2229.7 kJ/kg); all three are good at this low pressure but fail progressively towards the critical point.

WORKED EXAMPLE 4.11 — Effect of pressure on the melting point of ice

Problem. At 0°C the specific volumes of ice and water are 1.0907×10^{-3} and $1.0002 \times 10^{-3} \text{ m}^3/\text{kg}$ and the latent heat of fusion is 333.5 kJ/kg . Find the rate of change of the melting point of ice with pressure and the pressure needed to lower the melting point by 1 K . Why is the sign negative?

Given: $T = 273.15 \text{ K}$; $v_{sf} = v_f - v_s = 1.0002 \times 10^{-3} - 1.0907 \times 10^{-3} = -9.05 \times 10^{-5} \text{ m}^3/\text{kg}$; $h_{sf} = 333.5 \text{ kJ/kg}$.

Formula: $(dp/dT)_{\text{fus}} = h_{sf}/(T v_{sf})$; $(dT/dp)_{\text{fus}} = T v_{sf}/h_{sf}$.

Solution.

$(dp/dT)_{\text{fus}} = 333.5/(273.15 \times (-9.05 \times 10^{-5})) = 333.5/(-0.02472) = -13490 \text{ kPa/K} = -13.5 \text{ MPa/K}$.

$(dT/dp)_{\text{fus}} = 1/(-13490) = -7.41 \times 10^{-5} \text{ K/kPa} = -0.0074 \text{ K/bar}$.

Pressure to lower the melting point by 1 K : $\Delta p = 1 \times 13.5 = 13.5 \text{ MPa}$ (about 135 bar).

ANSWER The melting point of ice falls by 0.0074 K for every bar of pressure, and a pressure of 13.5 MPa lowers it by 1 K . The sign is negative because water *contracts* on melting ($v_{sf} < 0$): raising the pressure favours the denser phase, the liquid, so ice melts at a lower temperature – the fusion curve of water has a negative slope on the p – T diagram, unlike that of substances that expand on melting.

11. Practice problems (answers given)

No.	Problem	Answer
P1	Find the mass of carbon dioxide ($R = 0.1889 \text{ kJ/kg K}$) in a 2 m^3 tank at 500 kPa and 40°C , treating it as an ideal gas.	$m = 16.9 \text{ kg}$
P2	Steam at 5 MPa and 350°C has $v = 0.05197 \text{ m}^3/\text{kg}$ (tables). Find its compressibility factor ($R = 0.4615 \text{ kJ/kg K}$).	$Z = 0.904$
P3	Find the van der Waals constants of nitrogen ($T_c = 126.2 \text{ K}$, $p_c = 3.39 \text{ MPa}$, $R = 0.2968 \text{ kJ/kg K}$) and the volume of 1 kg at 300 K and 10 MPa by the ideal-gas and van der Waals equations.	$a = 0.1746 \text{ kPa m}^6/\text{kg}^2$, $b = 1.381 \times 10^{-3} \text{ m}^3/\text{kg}$; $v_{\text{ideal}} = 0.00890 \text{ m}^3/\text{kg}$; $v_{\text{vdW}} = 0.00858 \text{ m}^3/\text{kg}$ ($Z = 0.96$; measured $0.00895 \text{ m}^3/\text{kg}$, $Z = 1.005$ – near the Boyle temperature the corrections nearly cancel)
P4	Methane ($R = 0.5182 \text{ kJ/kg K}$, $T_c = 190.6 \text{ K}$, $p_c = 4.60 \text{ MPa}$) is at 8 MPa and 250 K . Find v by the ideal-gas equation and by the chart.	$p_R = 1.74$, $T_R = 1.31$, $Z \approx 0.74$; $v_{\text{ideal}} = 0.0162$, $v_{\text{chart}} = 0.0120 \text{ m}^3/\text{kg}$
P5	For mercury at 20°C , $\beta = 1.81 \times 10^{-4} \text{ K}^{-1}$, $\kappa_T = 3.9 \times 10^{-11} \text{ Pa}^{-1}$, $v = 7.39 \times 10^{-5} \text{ m}^3/\text{kg}$. Find $c_p - c_v$ ($c_p = 0.139 \text{ kJ/kg K}$).	0.0182 kJ/kg K (13% of c_p)
P6	Estimate the boiling point of water at 200 kPa from the Clausius–Clapeyron equation with $p = 101.42 \text{ kPa}$ at 100°C and $h_{fg} = 2257 \text{ kJ/kg}$.	120.4°C (tables: 120.2°C)
P7	Using the van der Waals low-pressure result $\mu_J \approx (2a/RT - b)/c_p$, estimate μ_J of nitrogen at 300 K ($c_p = 1.039 \text{ kJ/kg K}$; a , b from P3).	2.4 K/MPa (measured $\approx 2.1 \text{ K/MPa}$)

No.	Problem	Answer
P8	At the triple point of water (273.16 K, $v_g = 206.0 \text{ m}^3/\text{kg}$) $h_{fg} = 2501 \text{ kJ/kg}$ and $h_{sf} = 333.5 \text{ kJ/kg}$. Find the slopes of the vaporisation and sublimation curves.	0.0444 kPa/K and 0.0504 kPa/K – the sublimation curve is steeper

12. Summary and formula sheet

- Ideal gas:** $pV = RT$, $pV = mRT = nR_u T$; $R = R_u/M$, $R_u = 8.3145 \text{ kJ/kmol K}$; Boyle ($pV = C$ at $T = C$), Charles ($V/T = C$ at $p = C$), Avogadro ($22.414 \text{ m}^3/\text{kmol}$ at 0°C , 101.325 kPa). Assumptions: negligible molecular volume, no intermolecular forces, elastic collisions. $u, h = f(T)$; $c_p - c_v = R$.
- Real gas:** $Z = pv/(RT) \neq 1$; $Z < 1$ where attraction dominates, $Z > 1$ where molecular volume dominates; $Z \rightarrow 1$ as $p \rightarrow 0$ or at $T_R > 2$ (ideal-gas region $p_R \ll 1$); Boyle temperature: $(\partial Z/\partial p)_T \rightarrow 0$ as $p \rightarrow 0$ (vdW: $T_B = a/(Rb) = 27T_c/8$).
- Van der Waals:** $(p + a/v^2)(v - b) = RT$; $a = 27R^2T_c^2/(64p_c)$, $b = RT_c/(8p_c)$, $v_c = 3b$, $Z_c = 3/8$. **Redlich–Kwong:** $p = RT/(v - b) - a/[\sqrt{T} v(v + b)]$, $a = 0.42748 R^2T_c^{2.5}/p_c$, $b = 0.08664 RT_c/p_c$. **Beattie–Bridgeman:** $p = (R_u T/\bar{v}^2)(1 - c/\bar{v}^3)(\bar{v} + B) - A/\bar{v}^2$, $A = A_0(1 - a/\bar{v})$, $B = B_0(1 - b/\bar{v})$. **Virial:** $pV = RT(1 + B/V + C/V^2 + \dots)$.
- Reduced properties:** $p_R = p/p_c$, $T_R = T/T_c$, $v_R' = v/(RT_c/p_c) = ZT_R/p_R$. **Corresponding states:** $Z = f(p_R, T_R)$ for all gases $\rightarrow$ generalised chart; $v = ZRT/p$.
- Potentials:** $du = T ds - p dv$; $dh = T ds + v dp$; $df = -s dT - p dv$ ($f = u - Ts$); $dg = -s dT + v dp$ ($g = h - Ts$). **Maxwell:** $(\partial T/\partial v)_s = -(\partial p/\partial s)_v$; $(\partial T/\partial p)_s = (\partial v/\partial s)_p$; $(\partial p/\partial v)_T = (\partial s/\partial v)_T$; $(\partial v/\partial T)_p = -(\partial s/\partial p)_T$.
- Tds:** $T ds = c_v dT + T(\partial p/\partial T)_v dv = c_p dT - T(\partial v/\partial T)_p dp$. $c_p - c_v = T(\partial p/\partial T)_v(\partial v/\partial T)_p = -T(\partial v/\partial T)_p^2(\partial p/\partial v)_T = Tv\beta^2/\kappa_T \geq 0$ ($= R$ for an ideal gas, $\rightarrow 0$ at 0 K). $\gamma = c_p/c_v = \kappa_T/\kappa_s = (\partial p/\partial v)_s/(\partial p/\partial v)_T$. **Energy equation:** $(\partial u/\partial v)_T = T(\partial p/\partial T)_v - p$; $(\partial h/\partial p)_T = v - T(\partial v/\partial T)_p$; both zero for an ideal gas.
- Joule–Thomson:** $\mu_J = (\partial T/\partial p)_h = [T(\partial v/\partial T)_p - v]/c_p$; zero for an ideal gas; cooling inside the inversion curve; vdW $T_i = 2a/(Rb) = 27T_c/4$.
- Claapeyron:** $(dp/dT)_{\text{sat}} = h_{fg}/(T v_{fg})$ (any phase change; negative for the fusion of ice). **Clausius–Claapeyron:** $d(\ln p)/dT = h_{fg}/(RT^2)$, $\ln(p_2/p_1) = (h_{fg}/R)(1/T_1 - 1/T_2)$ ($v_f \ll v_g$, ideal vapour, h_{fg} constant).

13. Review questions

13.1 Short-answer (two-mark) questions

- Write the characteristic gas equation and define the universal gas constant.
- State Boyle's law.
- State Avogadro's law.
- What is an ideal gas? On what assumptions is the model based?
- How does a real gas differ from an ideal gas?
- Write the van der Waals equation and explain the two correction terms.
- Express the van der Waals constants a and b in terms of the critical pressure and temperature.
- Define the compressibility factor. What is its value for an ideal gas?
- What is the compressibility factor of a van der Waals gas at the critical point?
- Define reduced pressure, reduced temperature and (pseudo-)reduced volume.
- State the principle of corresponding states.
- How is the generalised compressibility chart used to find the properties of a real gas?
- Write the Beattie–Bridgeman equation and the virial equation of state.
- What is the Boyle temperature?
- Write the four Maxwell relations.
- Write the two Tds equations.
- Write the expression for $c_p - c_v$ in terms of the volume expansivity and the isothermal compressibility.

18. Express the ratio c_p/c_v in terms of the isothermal and isentropic compressibilities.
19. State Charles' law.
20. How do the universal and characteristic gas constants differ?
21. Define the Joule–Thomson coefficient. What is its value for an ideal gas?
22. Under what conditions does a real gas behave as an ideal gas?
23. Write the Clausius–Clapeyron equation and name the quantities it relates.
24. Define the Helmholtz function and the Gibbs function.
25. A gas ($R = 0.287 \text{ kJ/kg K}$) occupies 0.4 m^3 at 3 bar and 35°C ; find its mass.
26. What is the inversion curve? What happens to a gas throttled from a state outside it?
27. Why is c_p never less than c_v ?
28. Why does the melting point of ice fall as the pressure rises?
29. What is meant by the energy equation, and what does it give for an ideal gas?
30. Write the Redlich–Kwong equation of state.

13.2 Long-answer (thirteen-mark) questions

1. Derive the van der Waals equation from the ideal-gas equation by correcting for molecular volume and intermolecular attraction; using the critical-point conditions $(\partial p/\partial v)_T = 0$ and $(\partial^2 p/\partial v^2)_T = 0$, express a and b in terms of p_c and T_c . Write the Redlich–Kwong and Beattie–Bridgeman equations, explain their constants and compare the three equations.
2. Explain the generalised compressibility chart: define Z , p_R , T_R and v_R' , state the principle of corresponding states, describe the axes, the constant- T_R curves and the v_R' lines, identify the ideal-gas regions, and give the procedure for finding an unknown property from any two of p , T and v .
3. Find the specific volume of steam at 8 MPa and 400°C by the ideal-gas equation and by the compressibility chart, compare with the table value $0.03432 \text{ m}^3/\text{kg}$, evaluate the errors and explain them in terms of p_R and T_R ($R = 0.4615 \text{ kJ/kg K}$, $T_c = 647.1 \text{ K}$, $p_c = 22.06 \text{ MPa}$).
4. Nitrogen is stored at 8 MPa and 160 K in a 0.2 m^3 cylinder. Find the specific volume and mass by the ideal-gas equation and by the chart, and compare with the measured $0.003745 \text{ m}^3/\text{kg}$ ($R = 0.2968 \text{ kJ/kg K}$, $T_c = 126.2 \text{ K}$, $p_c = 3.39 \text{ MPa}$).
5. Water vapour at 400°C has $v = 0.020 \text{ m}^3/\text{kg}$. Find its pressure by the ideal-gas equation, the van der Waals equation (constants from the critical properties) and the compressibility chart with the pseudo-reduced volume; compare with the table value 12.52 MPa and comment.
6. A fluid at 250 kPa and 300°C in a 0.6 m^3 rigid vessel is cooled until the pressure is 125 kPa. For (i) air and (ii) steam find the mass, the final temperature and the heat transferred; show both processes on the T – v plane and explain the difference ($R = 0.287$, $c_v = 0.718 \text{ kJ/kg K}$ for air).
7. A fluid at 160°C with $v = 0.90 \text{ m}^3/\text{kg}$ expands at constant pressure to $v = 1.40 \text{ m}^3/\text{kg}$. For 1 kg of (i) air and (ii) steam find the pressure, the work, the heat transferred and the final temperature, and explain why the steam absorbs much more heat ($R = 0.287$, $c_p = 1.005 \text{ kJ/kg K}$ for air).
8. Starting from the first law and the Maxwell relations, derive the two Tds equations and apply them to obtain the entropy change of an ideal gas in terms of (T, v) and (T, p) .
9. Define the four thermodynamic potentials, write their exact differentials and derive the four Maxwell relations from them.
10. Derive the Clapeyron equation for a first-order phase change from a Maxwell relation, reduce it to the Clausius–Clapeyron equation for vaporisation at low pressure stating the assumptions, and use it to estimate the saturation pressure of water at 110°C from the value at 100°C .
11. By equating the two Tds equations derive $c_p - c_v = T v \beta^2 / \kappa_T$, and show that (i) $c_p - c_v = R$ for an ideal gas, (ii) c_p can never be less than c_v and (iii) $c_p = c_v$ at absolute zero. Evaluate $c_p - c_v$ for liquid water at 20°C .
12. Define the Joule–Thomson coefficient, describe the porous-plug experiment with the isenthalpic curves and inversion curve on the T – p plane, derive $\mu_J = (1/c_p)[T(\partial v/\partial T)_p - v]$, show that $\mu_J = 0$ for an ideal gas and discuss its sign and the inversion temperature for a real gas.

13. Derive the energy equation $(\partial u / \partial v)_T = T(\partial p / \partial T)_v - p$ and the corresponding enthalpy relation, and apply them to show that u and h of an ideal gas depend on temperature alone; find $(\partial u / \partial v)_T$ for a van der Waals gas.
14. Derive $\gamma = c_p / c_v = \kappa_T / \kappa_s$ and show that the slope of an isentrope on the p – v plane is γ times that of the isotherm through the same state; deduce $p v^\gamma = \text{constant}$ for an ideal gas.
15. Determine the van der Waals constants of carbon dioxide ($T_c = 304.2 \text{ K}$, $p_c = 7.38 \text{ MPa}$, $M = 44.01$) and estimate its pressure at 320 K and $0.004 \text{ m}^3/\text{kg}$ by the ideal-gas and van der Waals equations, comparing with the measured 8.27 MPa .
16. Estimate the enthalpy of vaporisation of water at 100°C from the saturation pressures at 95°C and 105°C using the Clapeyron equation, and find the pressure needed to lower the melting point of ice by 1 K (data as in Worked Examples 4.9 and 4.11).

Text: P. K. Nag, "Engineering Thermodynamics", 6th Edn., McGraw-Hill — Chapters 10–11. *References:* Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.; R. K. Rajput, "A Text Book of Engineering Thermodynamics", 5th Edn. *Real-gas data:* IAPWS-97, NIST.